

convert it into a floating-point value this value is multiplied by a fixed-point 24-bit representation of 0.1 to obtain a 48-bit floating-point value of time in seconds. This is going to get a bit heavy but we promise you it will make sense.

This failure was caused by the 24-bit representation of 0.1 in base 2. So the base 2 representation of 0.1 is a non-terminating 0.00011001100110011001100... The 24-bit register in the Patriot instead stored a terminating 0.00011001100110011001100. Because of this switch from terminating to non-terminating decimal, an error of 0.000000000 00000000000000011001100... binary, or about 0.000000095 decimal was introduced resulting in the accuracy of the time being reduced by 0.0001%. Now this is a very small time difference and is almost insignificant, especially because the tracking of the missile does not depend on the absolute clock-time, but in fact on the time elapsed between two different radar pulses. But there was another issue.

The software which was used by the computer was written by the assembly language more than 20 years ago. To stay relevant to the high speed of ballistic missiles it was updated several times, but unfortunately, the system was not originally designed for handling this data. One of these updates contained a feature to convert clock-time to floating point with more precision. So now a weird thing happened. You have one less accurate time of one radar pulse being subtracted from a more accurate time of another radar pulse. If both these times suffered the same amount of inaccuracy, the error would have cancelled itself out. This was not the case here though.

So when the Scud missile arrived, this computer had been active for 100 hours. Remember that we had an error of 0.000000095 decimal from one radar pulse. So to find out the error, $0.000000095 \times 100 \times 60 \times 60$, which is further multiplied by 10 because time was stored in values of one tenth of a second. So effectively, the calculated time elapsed had an error of 0.34 seconds. Since the Scud missile travels at about 1680 meters per second it travels more than half a kilometer in this time. Half a kilometer is enough to be outside the 'range gate'



Patriot Missile Failure



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